URBANSTYLE.LTD

Data Quality Investigation

Team Report — Week 1 Group Session

Roles A–D · sales · customers · products · channels & locations

THE CHALLENGE

"I don’t trust this data yet… I want a separate report on EVERY table." — Toomas Kask, IT Director

His Week 1 letter had already flagged sales-data concerns; in the live group session he widened the ask to every table. The team split the work by role, one table per person, using only Week 1 skills.

OUR APPROACH — FOUR ROLES, FOUR TABLES

Week 1 toolkit only — SELECT, WHERE, COUNT, DISTINCT, ORDER BY, LIMIT. Read-only: nothing in the database was changed.



ROLE A

Sales transactions

table: sales



ROLE B

Customer data

table: customers



ROLE C

Product catalogue

table: products



ROLE D

Channels & locations

table: sales (channel/location fields)

ROLE A — DARIA

Sales transactions

"I explored the sales transactions, focusing mainly on amounts, dates, and data quality."

Transactions: 15,234, covering January 2023 to June 2026

Transaction values: range from −€1,405.32 to €2,170.40

I found two things that I think are particularly important:

9.8% missing a customer ID

1,487 transactions, or 9.8%, don't have a customer ID, so we can't connect those sales to individual customers.

2% have a negative total price

305 transactions, around 2% of the dataset, have negative total prices. We don't yet know what those transactions represent. They could potentially be returns, refunds, or corrections, but we would need more information before making that conclusion.

ROLE B — REIO

Customer data

I investigated the customers table and found 3,150 customers total, with data that's clean on names.

Customers total: 3,150

Missing first names: 0 — clean on names

Missing emails: 380 customers (12%)

Registration dates: run from January 2020 to February 2025

The standout finding was city

54 different spellings for what should be about a dozen real cities — Tallinn alone is split across at least 4 variants from casing and stray spaces — which means any report grouping by city right now will quietly undercount rather than error out.

ROLE C — MARIA

Product catalogue

I investigated the products table and found 362 products across five distinct categories.

Products: 362, across five distinct categories

Retail prices: ranged from €13.53 to €434.08, showing a fairly wide product price range

No missing retail price or category values

I also checked for missing values and found no missing retail price or category values. This means the product data appears complete for basic price and category analysis, although further checks would be needed before concluding that the whole products table is free of data-quality issues.

ROLE D — MARVIS

Sales channels & locations

I investigated the sales channels and locations.

Channels identified: Online and shop

Unique store locations: Tallinn, Pärnu, Tartu, and Null

Payment methods: Cash, Card, and installments

Tallinn: Card, Cash

Pärnu: installments

Tartu: Cash, Card, installments

Online transactions: made through online stores show a consistent quantity of 5

What Was Surprising / Key Anomalies

Missing Store Locations: 10,408 transactions recorded with no store location attached.

What We Learned — and What's Next

PATTERNS ACROSS ALL FOUR TABLES



Gaps, not broken links

Every FK checks out (0 orphaned customer IDs) — the gaps are optional fields left blank: customer ID, store location, email.



Free text needs input rules

City names alone came back in 54 spellings — nothing at entry enforces a controlled list.



Some patterns need a business answer

Negative prices and a constant online quantity of 5 both look intentional — only Toomas's team can say what they mean.

OPEN QUESTIONS FOR TOOMAS

1. Are negative-price transactions returns, refunds, or errors?
2. Is the “Null” store location a system default, or truly missing data?
3. Should city values be standardized at entry, or cleaned downstream?
4. What definition of “duplicate sale” should Week 2 validate against?

Questions?

UrbanStyle.ltd Week 1 Group Session — the team behind this report

A

Daria

Sales

B

Reio

Customers

C

Maria

Products

D

Marvis

Channels & Locations

Presented by Reio on behalf of the team